

Generation Interconnection System Impact Study Report

For Queue Project Q19-1348

Moreno Tap 230 kV

October 27, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1348. The Interconnection Customer proposes a 112.9 MW solar photovoltaic facility interconnecting at the Moreno Tap 230 kV substation in the Ohio area. The request was received on December 04, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 525 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$101,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Little Chute Substation – Talenenergy Martins Crk Llc Harrisburg 345kV	Fort Humbug – Coronet 500kV Line	102.1	2,249	6.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Little Chute Substation – Talenenergy Martins Crk Llc Harrisburg 345kV	Reconductor line	\$10,728,000	1% (\$101,000)
	Total allocated to project		\$101,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.